Projekt neve: Course Management System
Technológia: Java 17, Spring Boot
Adatbázis: MySQL (éles) + H2 (teszt)
Tesztelés: JUnit 5 + Mockito

Cél

Egy REST API backend alkalmazás, ami kezeli kurzusokat, felhasználókat és kategóriákat, és biztosít CRUD műveleteket, valamint hallgatók kurzusra való feliratkozását.

```
src/main/java/com/example/course_management
 ├─ controller
 │    ├ CourseController.java
 │    ├ UserController.java
 │    └ CategoryController.java
 ├─ service
 │    ├ CourseService.java
 │    ├ UserService.java
 │    └ CategoryService.java
 ├─ repository
 │    ├ CourseRepository.java
 │    ├ UserRepository.java
 │    └ CategoryRepository.java
 ├─ entity
 │    ├ Course.java
 │    ├ User.java
 │    └ Category.java
 ├─ dto
```

```
│   ├ CourseRequestDto.java
│   ├ CourseResponseDto.java
│   ├ UserRequestDto.java
│   ├ UserResponseDto.java
│   ├ CategoryRequestDto.java
│   └ CategoryResponseDto.java
└ mapper
    CourseMapper.java
```

REST API Lista

HTTP	Endpoint	Leírás
GET	/api/courses	Minden kurzus
GET	/api/courses/{id}	Kurzus ID alapján
POST	/api/courses	Kurzus létrehozása
PUT	/api/courses/{id}	Kurzus módosítása
DELETE	/api/courses/{id}	Kurzus törlése
POST	/api/courses/{courseId}/enroll/{studentId}	Hallgató feliratkozása
POST	/api/courses/{courseId}/categories/{categoryId}	Kategória hozzárendelés

DTO minták

CourseRequestDto

```
{
  "title": "Spring Boot Advanced",
  "description": "Advanced backend development",
  "maxParticipants": 20,
  "startDate": "2026-05-01",
  "instructorId": 1
}
```

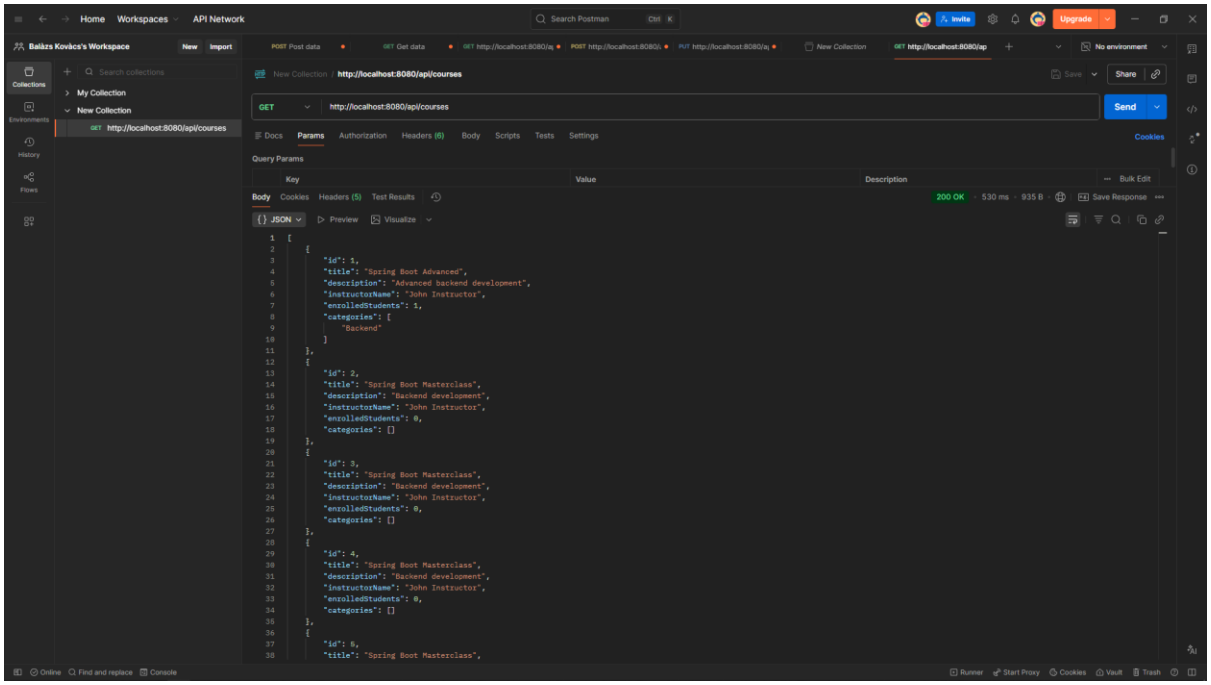
UserRequestDto

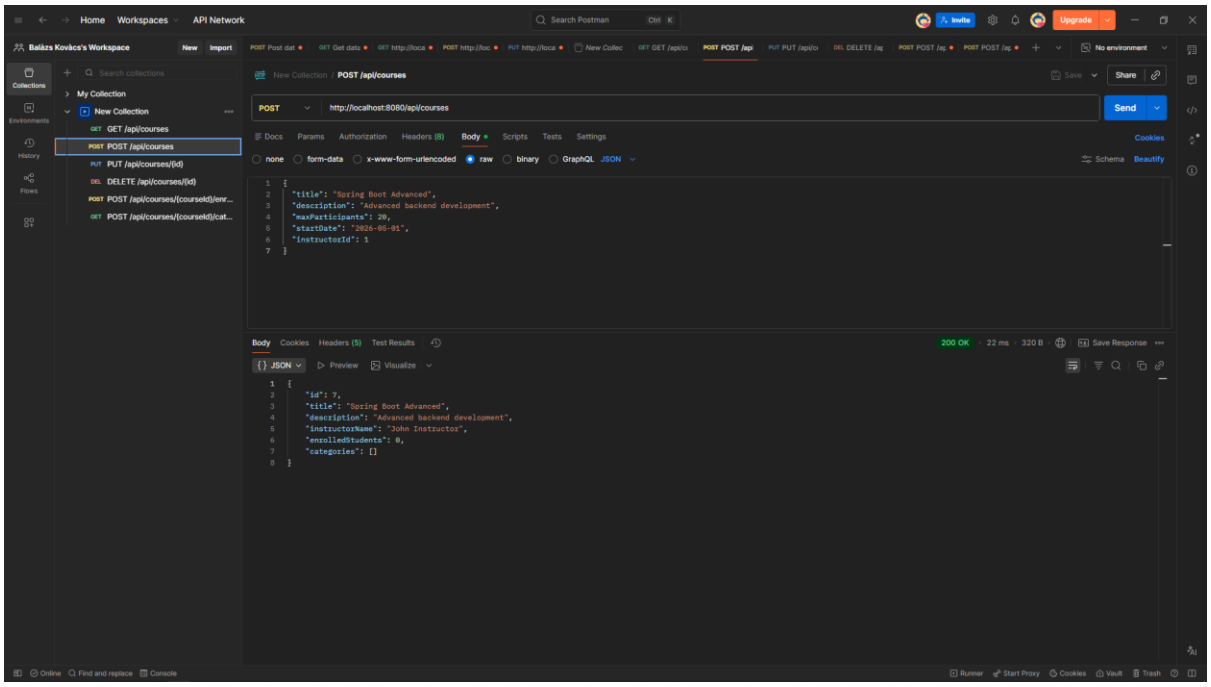
```
{
  "name": "John Instructor",
  "email": "john@example.com",
  "password": "1234",
  "role": "INSTRUCTOR"
}
```

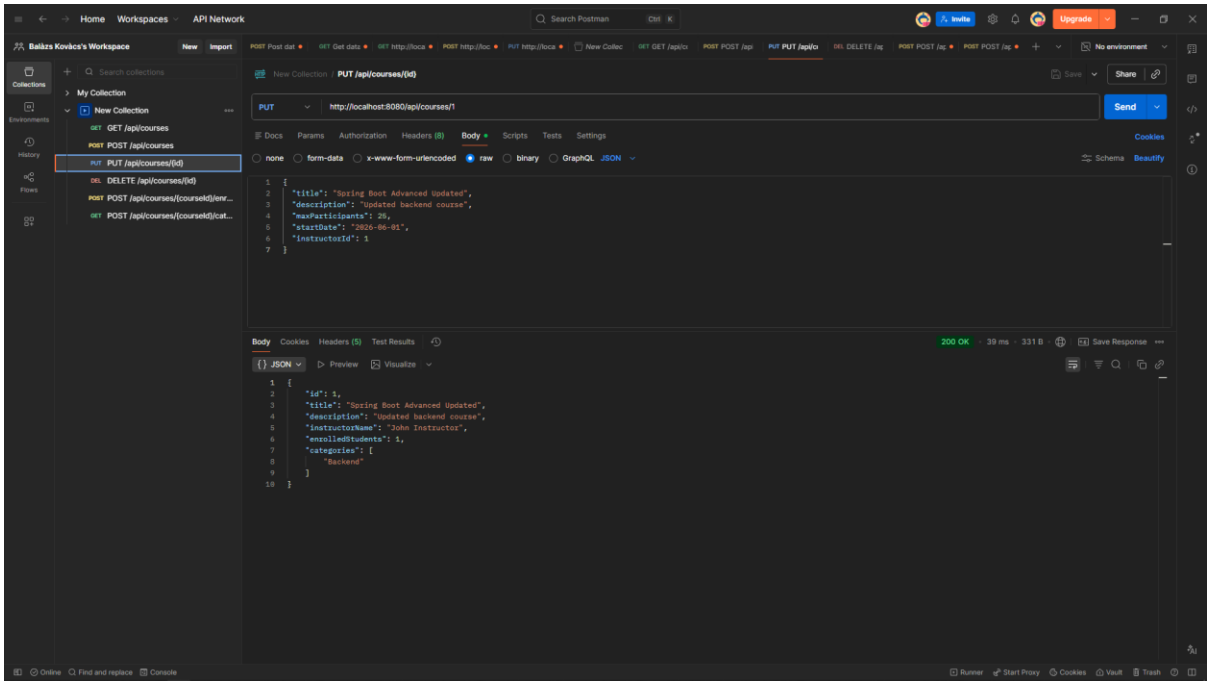
CategoryRequestDto

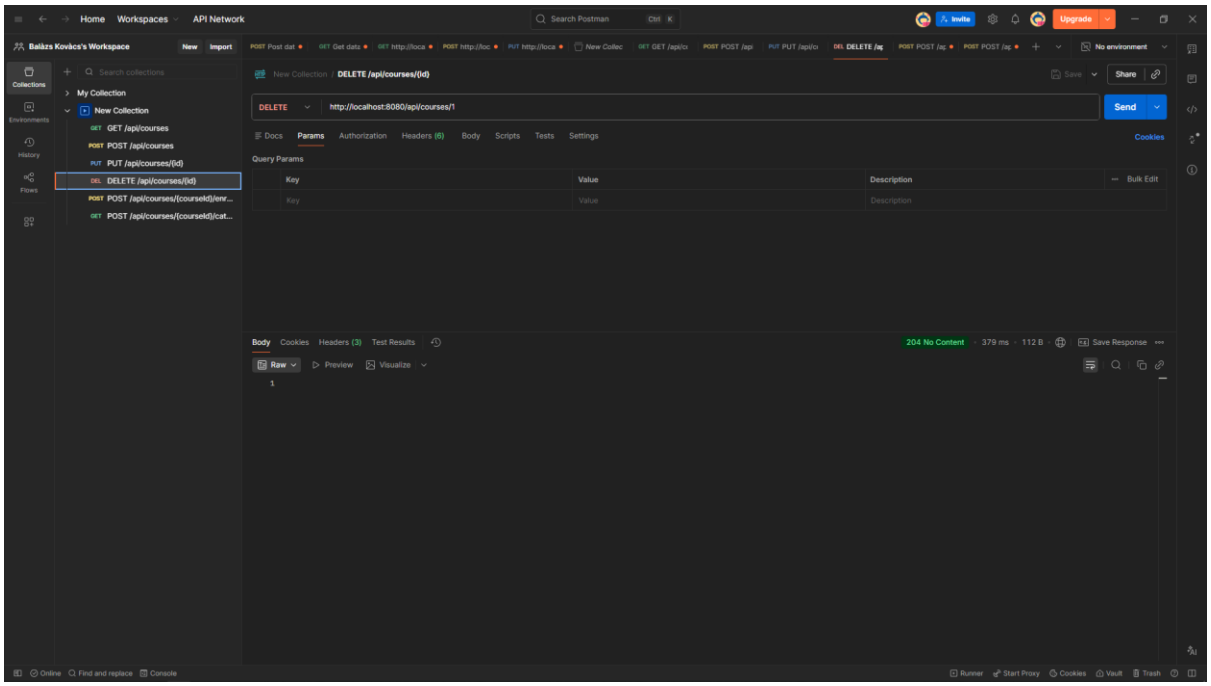
```
{
  "name": "Backend",
  "code": "BE",
  "description": "Backend courses",
  "active": true
}
```

Postman teszt képek






Home
Workspaces
API Network
Search Postman
Invite
Upgrade
Balázs Kovács's Workspace
New
Import
Collections
Environments
History
Flows
My Collection
New Collection
GET /api/courses
POST POST /api/courses
PUT /api/courses/{id}
DEL DELETE /api/courses/{id}
POST POST /api/courses/{courseId}/enr...
GET POST /api/courses/{courseId}/cat...
New Collection / PUT /api/courses/{id}
Save
Share
PUT
http://localhost:8080/api/courses/1
Send
Docs
Params
Authorization
Headers (8)
Body
Scripts
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
{
"title": "Spring Boot Advanced Updated",
"description": "Updated backend course",
"maxParticipants": 20,
"startDate": "2026-06-01",
"instructorId": 1
}
Body
Cookies
Headers (5)
Test Results
200 OK
39 ms
331 B
Save Response
JSON
Preview
Visualize
{
"id": 1,
"title": "Spring Boot Advanced Updated",
"description": "Updated backend course",
"instructorName": "John Instructor",
"enrolledStudents": 1,
"categories": [
"Backend"
]
}
Online
Find and replace
Console
Runner
Start Proxy
Cookies
Vault
Trash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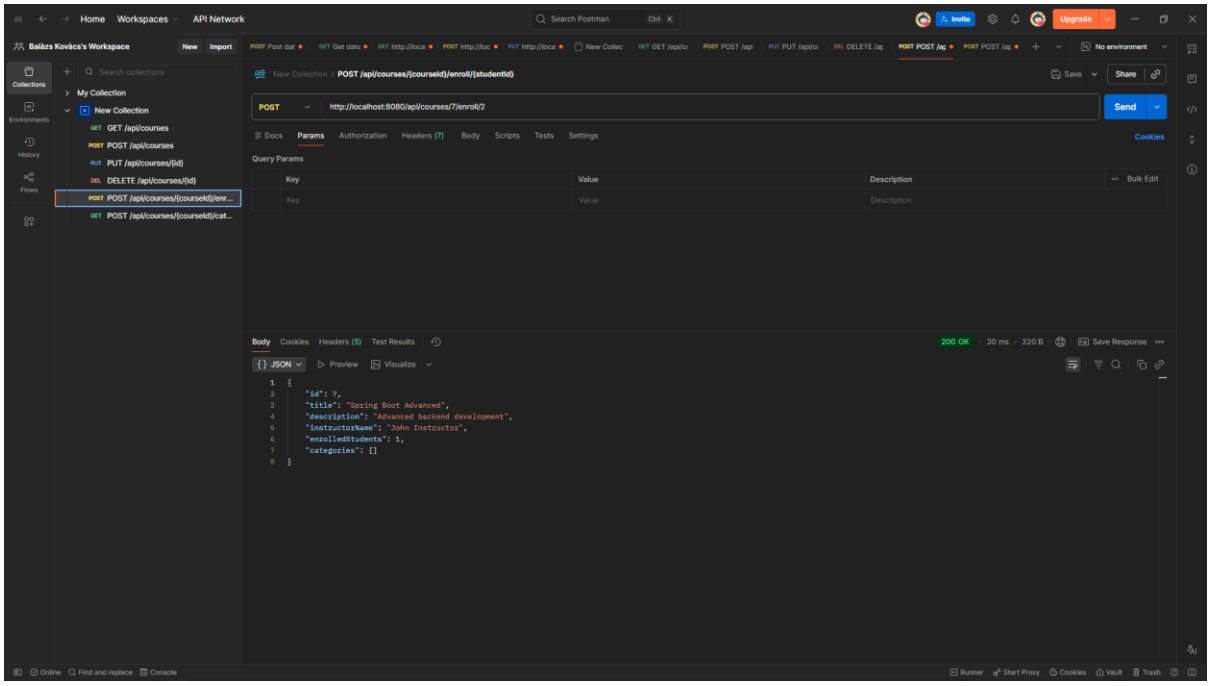
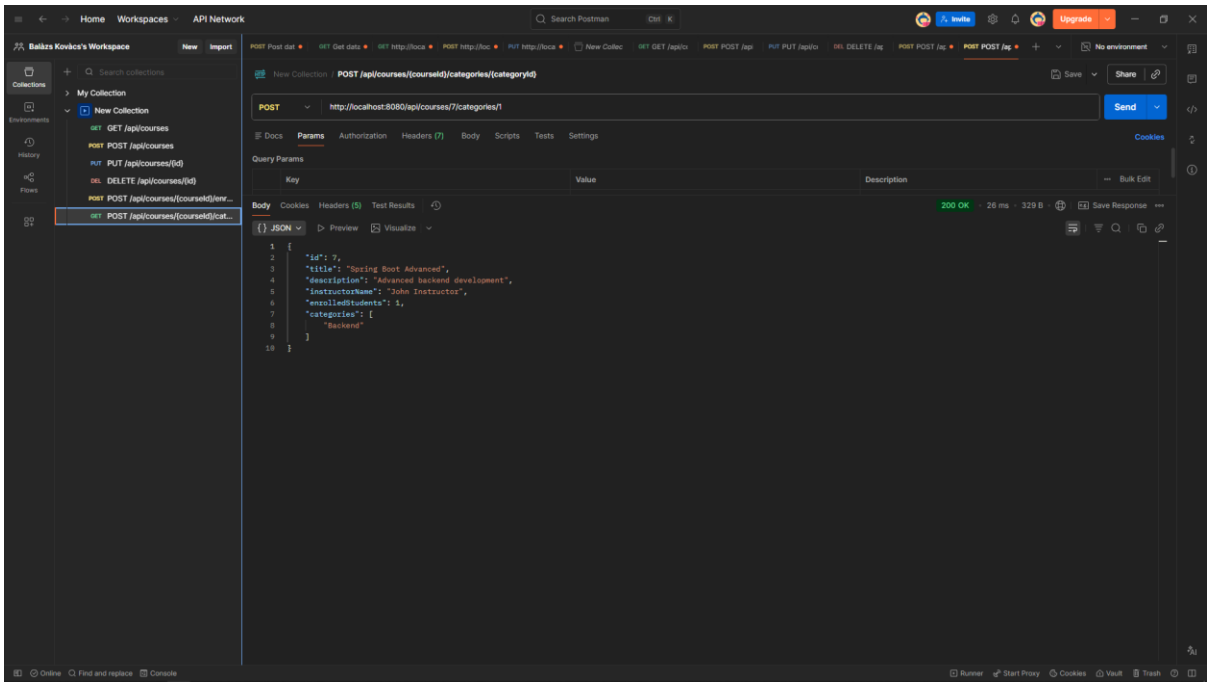
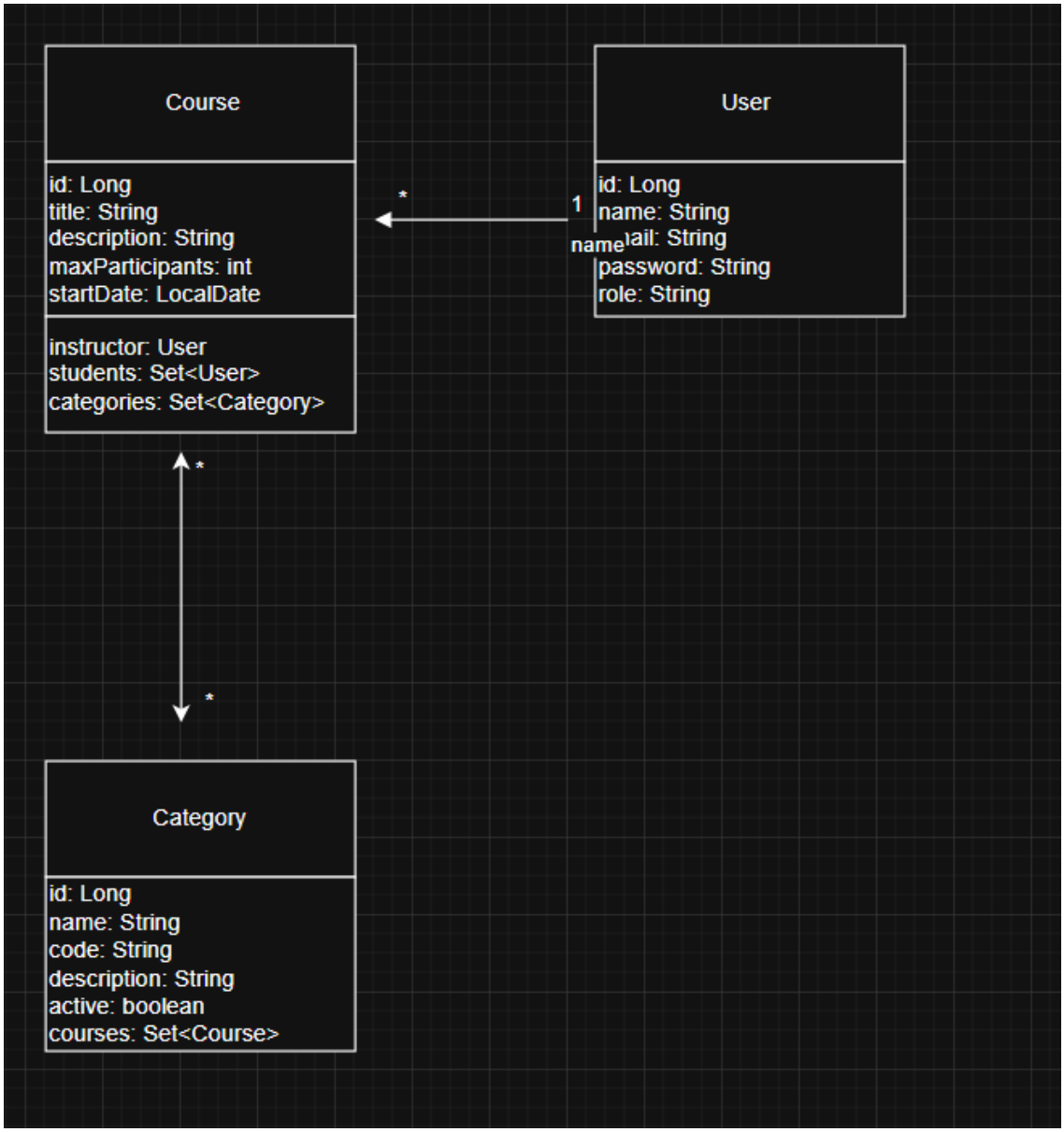
PUT /api/courses/{id}
DEL DELETE /api/courses/{id}
POST POST /api/courses/{courseId}/enr...
GET POST /api/courses/{courseId}/cat...
New Collection / DELETE /api/courses/{id}
Save
Share
DELETE
http://localhost:8080/api/courses/1
Send
Docs
Params
Authorization
Headers (6)
Body
Scripts
Tests
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Body
Cookies
Headers (3)
Test Results
204 No Content
379 ms
112 B
Save Response
Raw
Preview
Visualize
Online
Find and replace
Console
Runner
Start Proxy
Cookies
Vault
Trash





UML Diagramok

Class Diagram



Course 1 ----- 1 User (instructor)

Course * ----- * User (students)

Course * ----- * Category

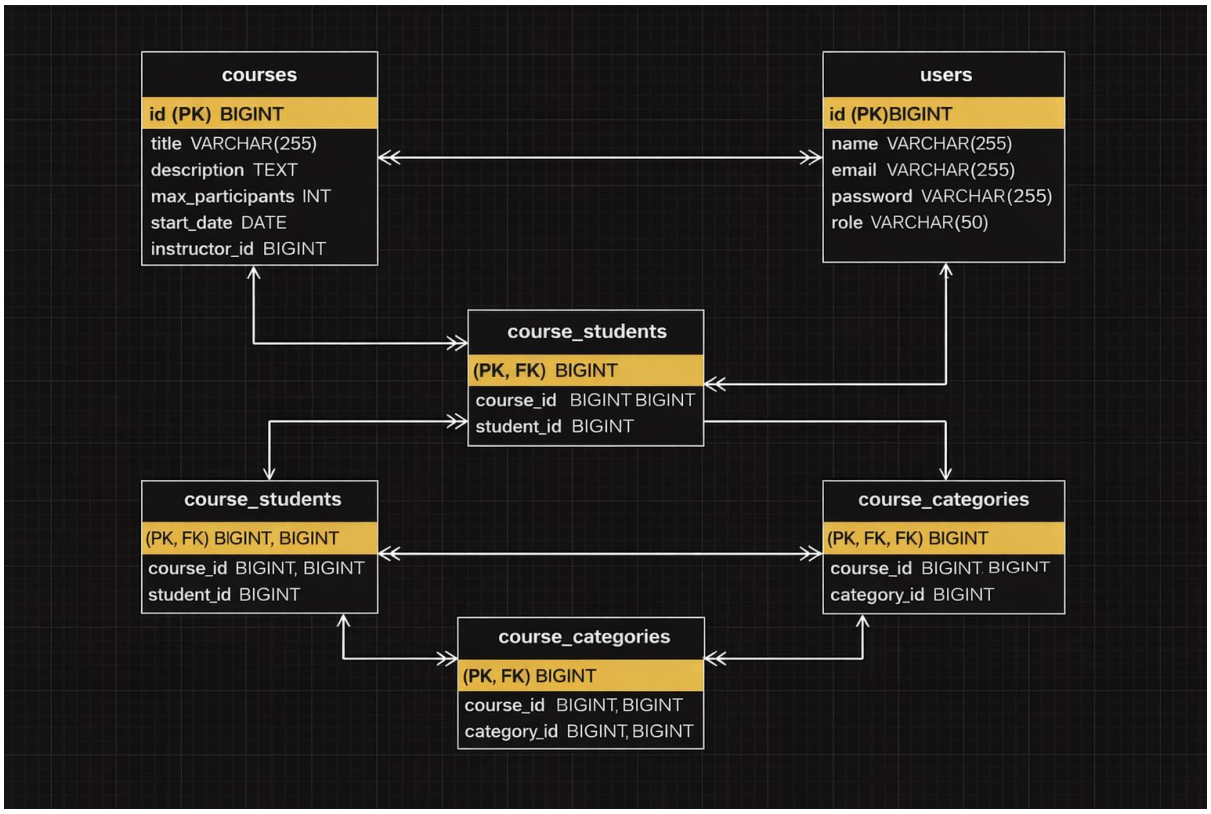
Sequence Diagram

```
Client (Postman)
|
| POST /api/courses
v
CourseController
|
| createCourse(dto)
v
CourseService
|
| find instructor by id
v
UserRepository
|
| returns instructor
v
CourseService
|
| save(course)
v
CourseRepository
|
| returns saved course
v
CourseService
|
| map to ResponseDto
v
CourseController
|
| return ResponseEntity
v
Client (Postman)
```

Category hozzárendelés:

```
Client (Postman)
|
| POST /api/courses/{courseId}/categories/{categoryId}
v
CourseController
|
| addCategoryToCourse(courseId, categoryId)
v
CourseService
|
| find course by id
| find category by id
| course.getCategories().add(category)
| save(course)
v
CourseRepository
|
| returns updated course
v
CourseService
|
| map to ResponseDto
v
CourseController
|
| return ResponseEntity
v
Client (Postman)
```

Adatbázis (ER) Diagram



Tesztelés

Unit test: CourseServiceTest (Mockito + JUnit5)

Validation: @NotBlank, @NotNull, @Email a DTO-kban

Exception kezelés: ResourceNotFoundException